

## IVANHOE, Australia

WMO: 956970

Lat: 32.8831S

Lon: 144.3089E

Elev: 330

StdP: 14.52

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 36.0       | 38.0 | 16.4                          | 12.7 | 77.3 | 19.7 | 15.0 | 74.6 | 21.2                  | 60.6 | 19.7 | 60.5 | 6.5                   | 210  | 0.500 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 24.7                   | 106.2           | 67.0 | 102.6 | 66.3 | 99.2 | 65.4 | 73.0                | 86.8 | 71.3 | 86.7 | 69.9 | 87.6 | 11.5                 | 340  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 70.3                            | 113.0 | 75.6 | 67.3 | 101.7 | 75.2 | 64.7 | 92.6 | 75.6 | 36.9          | 86.8 | 35.4 | 85.5 | 34.0 | 87.5 | 79.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 23.4              | 20.1 | 17.7 | 31.8                       | 110.3 | 1.9                | 3.5 | 30.4                                               | 112.8 | 29.3       | 114.9 | 28.2       | 116.8 | 26.9       | 119.4 |
|                   |      |      | 30.7                       | 76.1  | 1.9                | 1.9 | 29.3                                               | 77.5  | 28.2       | 78.6  | 27.1       | 79.7  | 25.7       | 81.1  |
|                   |      |      | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 66.4        | 82.6  | 80.5  | 74.0  | 66.6  | 57.8  | 52.4  | 51.7  | 53.8  | 60.2  | 67.2  | 73.3  | 77.8  | (6)  |
|                                                                        | DBStd       | 12.69       | 7.95  | 7.86  | 7.03  | 6.73  | 5.39  | 4.49  | 4.50  | 4.92  | 6.46  | 7.96  | 8.43  | 8.01  | (7)  |
|                                                                        | HDD50       | 72          | 0     | 0     | 0     | 0     | 2     | 25    | 32    | 13    | 0     | 0     | 0     | 0     | (8)  |
|                                                                        | HDD65       | 1701        | 0     | 0     | 9     | 59    | 230   | 379   | 414   | 348   | 175   | 68    | 16    | 3     | (9)  |
|                                                                        | CDD50       | 6064        | 1010  | 855   | 745   | 498   | 245   | 96    | 84    | 132   | 306   | 532   | 698   | 863   | (10) |
|                                                                        | CDD65       | 2221        | 545   | 436   | 289   | 107   | 8     | 0     | 0     | 3     | 31    | 136   | 264   | 402   | (11) |
|                                                                        | CDH74       | 29844       | 7860  | 5849  | 3327  | 1119  | 75    | 1     | 2     | 35    | 497   | 1876  | 3607  | 5596  | (12) |
|                                                                        | CDH80       | 16303       | 4863  | 3389  | 1578  | 376   | 3     | 0     | 0     | 5     | 148   | 848   | 1908  | 3185  | (13) |
| (14) Wind                                                              | WSAvg       | 9.3         | 10.5  | 10.3  | 9.6   | 8.3   | 8.1   | 8.0   | 8.2   | 8.7   | 9.4   | 10.2  | 10.3  | 10.4  | (14) |
| (15) Precipitation                                                     | PrecAvg     | 12.9        | 1.8   | 0.7   | 1.3   | 1.1   | 1.4   | 0.8   | 0.9   | 1.1   | 0.9   | 1.1   | 0.9   | 1.0   | (15) |
|                                                                        | PrecMax     | 34.6        | 8.3   | 4.8   | 7.4   | 7.0   | 3.8   | 2.5   | 2.7   | 2.7   | 3.2   | 3.9   | 4.7   | 4.1   | (16) |
|                                                                        | PrecMin     | 2.7         | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.1   | 0.0   | 0.0   | 0.0   | (17) |
|                                                                        | PrecStd     | 5.9         | 2.1   | 1.1   | 1.6   | 1.5   | 1.1   | 0.7   | 0.7   | 0.8   | 0.8   | 1.1   | 0.9   | 1.0   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 111.2 | 110.1 | 100.4 | 93.7  | 79.6  | 71.7  | 72.7  | 78.7  | 89.3  | 97.7  | 105.5 | 107.2 | (19) |
|                                                                        |             | MCWB        | 68.1  | 70.6  | 65.5  | 63.9  | 59.1  | 58.1  | 55.7  | 55.8  | 57.8  | 61.4  | 65.1  | 65.5  | (20) |
|                                                                        | 2%          | DB          | 106.9 | 104.9 | 95.8  | 87.8  | 76.0  | 66.9  | 68.2  | 72.9  | 84.2  | 93.4  | 99.8  | 102.6 | (21) |
|                                                                        |             | MCWB        | 67.5  | 67.6  | 64.7  | 62.2  | 58.5  | 55.1  | 53.8  | 53.8  | 57.2  | 60.7  | 63.9  | 65.4  | (22) |
|                                                                        | 5%          | DB          | 103.3 | 100.1 | 92.4  | 84.4  | 72.5  | 64.1  | 64.3  | 69.0  | 79.8  | 88.9  | 94.9  | 99.0  | (23) |
|                                                                        |             | MCWB        | 67.2  | 66.3  | 64.3  | 61.0  | 57.0  | 54.0  | 52.0  | 52.2  | 55.8  | 59.8  | 63.1  | 64.3  | (24) |
|                                                                        | 10%         | DB          | 99.3  | 95.8  | 88.8  | 80.6  | 68.9  | 61.3  | 61.5  | 65.5  | 75.2  | 84.5  | 90.8  | 95.1  | (25) |
|                                                                        |             | MCWB        | 66.3  | 66.2  | 63.8  | 59.7  | 55.8  | 52.9  | 51.2  | 51.2  | 54.3  | 58.8  | 61.2  | 63.7  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 75.2  | 75.9  | 73.0  | 67.9  | 64.3  | 60.1  | 61.7  | 59.1  | 61.9  | 66.8  | 70.7  | 73.9  | (27) |
|                                                                        |             | MCDB        | 86.6  | 87.7  | 82.3  | 79.0  | 68.0  | 65.0  | 64.9  | 66.9  | 75.3  | 82.4  | 89.0  | 82.9  | (28) |
|                                                                        | 2%          | WB          | 72.4  | 73.4  | 70.6  | 65.4  | 61.1  | 57.6  | 56.5  | 56.4  | 59.6  | 64.4  | 68.6  | 70.7  | (29) |
|                                                                        |             | MCDB        | 88.5  | 89.2  | 81.5  | 76.2  | 70.6  | 62.8  | 62.9  | 68.1  | 75.7  | 82.1  | 82.9  | 85.3  | (30) |
|                                                                        | 5%          | WB          | 70.9  | 71.5  | 68.2  | 63.6  | 59.0  | 56.1  | 53.8  | 54.0  | 57.6  | 62.5  | 66.7  | 68.6  | (31) |
|                                                                        |             | MCDB        | 89.9  | 85.4  | 82.3  | 76.6  | 68.4  | 61.3  | 61.3  | 65.8  | 74.2  | 81.5  | 82.7  | 87.2  | (32) |
|                                                                        | 10%         | WB          | 69.6  | 69.6  | 66.3  | 61.9  | 57.2  | 54.5  | 52.2  | 52.3  | 55.8  | 60.6  | 64.7  | 66.6  | (33) |
|                                                                        |             | MCDB        | 90.5  | 87.7  | 82.6  | 77.0  | 66.6  | 59.7  | 59.9  | 63.2  | 72.5  | 80.7  | 85.9  | 88.1  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 24.7  | 23.2  | 22.8  | 23.0  | 20.1  | 17.6  | 19.1  | 21.2  | 24.3  | 25.5  | 25.0  | 25.1  | (35) |
|                                                                        |             | MCDBR       | 28.4  | 27.4  | 26.3  | 26.5  | 23.9  | 21.8  | 23.0  | 26.7  | 30.5  | 29.9  | 29.1  | 28.6  | (36) |
|                                                                        | 5% WB       | MCWBR       | 9.6   | 9.6   | 10.1  | 11.2  | 11.3  | 11.8  | 12.3  | 12.7  | 12.6  | 12.0  | 10.1  | 10.1  | (37) |
|                                                                        |             | MCWBR       | 23.0  | 19.9  | 20.8  | 20.9  | 18.5  | 16.4  | 19.1  | 22.2  | 25.1  | 24.8  | 23.0  | 22.3  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.332 | 0.335 | 0.317 | 0.304 | 0.292 | 0.287 | 0.286 | 0.294 | 0.329 | 0.326 | 0.340 | 0.331 | (40) |
|                                                                        |             | taud        | 2.530 | 2.549 | 2.581 | 2.586 | 2.590 | 2.597 | 2.586 | 2.552 | 2.438 | 2.471 | 2.471 | 2.504 | (41) |
|                                                                        | Ebn at Noon |             | 319   | 312   | 307   | 294   | 282   | 275   | 281   | 293   | 298   | 311   | 314   | 320   | (42) |
|                                                                        |             | Edh at Noon | 35    | 33    | 30    | 27    | 23    | 22    | 23    | 27    | 34    | 36    | 37    | 36    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 2516  | 2241  | 1909  | 1460  | 1065  | 858   | 950   | 1262  | 1712  | 2112  | 2324  | 2543  | (44) |
|                                                                        | RadStd      |             | 144   | 117   | 115   | 71    | 62    | 49    | 51    | 83    | 96    | 122   | 165   | 96    | (45) |

## Historical Trends

|                             |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|-----------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                             |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (2 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page